

① 255.375

to binary:

$$\begin{array}{r|l} 2 & 255 \\ \hline 2 & 127 \quad 1 \\ 2 & 63 \quad 1 \\ 2 & 31 \quad 1 \\ 2 & 15 \quad 1 \\ 2 & 7 \quad 1 \\ 2 & 3 \quad 1 \\ 2 & 1 \quad 1 \\ & 0 \quad 1 \end{array}$$

~~$$\begin{array}{r|l} 2 & 375 \\ \hline 2 & 187 \quad 1 \\ 2 & 93 \quad 1 \\ 2 & 46 \quad 1 \\ & 23 \quad 0 \end{array}$$~~

$$0.375 \times 2 = 0.75 \quad 0$$

$$0.75 \times 2 = 1.5 \quad 1$$

$$0.5 \times 2 = 1 \quad 1$$

$$\therefore (11111111.011)_2$$

to Octal -

8	255	
8	31	7
8	3	7
	0	3

~~0.375~~

~~|||||~~

.011
→ 0.3

377.3

to hexadecimal -

|||| |
→ F → F

0.11
→ 0.6

FF.6

② 110101.101

to decimal

$$\Rightarrow 1 \cdot (2^0 \times 1) + (2^1 \times 1) + (2^2 \times 0) + (2^3 \times 0) + (2^4 \times 1) + 2^5 \times 1$$

$$\Rightarrow 1 + 2 + 0 + 0 + 16 + 32$$

$$\Rightarrow 53$$

$$\Rightarrow .101_2 = \left(\frac{1}{2} \times 1\right) + \left(\frac{1}{4} \times 0\right) + \left(\frac{1}{8} \times 1\right)$$

$$= \frac{1}{2} + 0 + \frac{1}{8} = \frac{5}{8} = 0.625$$

53.625

③ Divisibility test for binary³ⁱⁿ -

→ if difference in sum of bits at alternate positions is divisible by 3.

→ 100111

→ $(1+0+1) - (0+1+1)$

→ $2 - 2$

→ 0

∴ divisible by 3.

⑤ $A+B$ stands for OR

$$A+C \rightarrow A' = !A \quad A' \text{ OR } C$$

$$B+C' \rightarrow B \text{ OR } !C$$

$()()$ multiplication of 2 brackets stands for AND

→ $F = (A \parallel B) \Delta (!A \parallel C) \Delta (B \parallel !C)$

$F=1$ if $A, B, C=1$ or $B, C=1, A=0$

$F=0$ for everything else

→ ~~$F = A \Delta B \Delta C$~~

$F = B \Delta C$

⑥ 1, 3, 5, 7

K-map

	A	B	C
1	0	0	1
3	0	1	1
5	1	0	1
7	1	1	1

$C=1$ for all min terms

$\Rightarrow F=C$

④ -23

$23 = (\cancel{0111})_2 \quad (00010111)_2$

to make -ve, take complement

$-23 = (\cancel{01000})_2 \quad (11101000)_2$